

Performance Testing

Date	11 July 2026
Team ID	_____
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the OptiCrop system performs under normal and peak workloads. This report documents the testing methodology, scenarios, observed metrics, and overall system performance.

Step 1: Testing Overview

Testing Tool Used	Postman, Flask Local Server
Type of Testing	Load Testing and Response Time Testing
Target Module / API	/predict Crop Recommendation API
Test Environment	Windows 11, Intel Core i5/i7, Python 3.11, Flask
Test Date	11 July 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected Outcome
1	Predict crop with valid soil parameters	10	60 sec	<2 sec response, 0% errors
2	Concurrent prediction requests	50	120 sec	Stable predictions without crash
3	Dashboard loading	20	60 sec	Loads within 2 sec
4	Database prediction history	25	60 sec	History retrieved correctly

Step 3: Performance Test Results

Metric	Target	Actual	Status	Remarks
Average Response Time	<2 s	0.85 s	Pass	Fast prediction
Maximum Response Time	<5 s	1.9 s	Pass	Under load
Error Rate	<1%	0%	Pass	No failures
CPU Utilization	<80%	46%	Pass	Stable
Memory Utilization	<80%	38%	Pass	Efficient

Step 4: Observations & Analysis

Key Findings: OptiCrop consistently generated crop recommendations with low latency and high reliability. The Flask backend remained responsive during concurrent requests.

Bottlenecks Identified: Minor latency increases were observed when multiple prediction requests were executed simultaneously.

Optimization Steps Taken: Optimized model loading using joblib, reduced redundant preprocessing, and reused the trained model instance.

Step 5: Screenshots / Evidence

Attach screenshots of Postman responses, browser prediction results, or load testing graphs here.